MILITARY SPECIFICATION

PESTICIDE DISPERSAL UNIT (PDU),

MULTICAPABILITY, HELICOPTER SLUNG

This specification is approved for use by all Departments and and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a Pesticide Dispersal Unit (PDU), Multicapability, Helicopter Slung, capable of applying pesticide formulations in two different liquid modes, low volume (LV) and ultra low volume (ULV), and a solid dispersal (SD) mode for granular pesticide dispersal. The PDU is independent of the aircraft for electrical or mechanical power requirements.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Biomedical Research and Development Laboratory, ATTN: SGRD-UBE-D, Fort Detrick, Frederick, MD 21701-5010, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3740

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

ENCL 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

2.1.1 (cont'd)

SPECIFICATIONS

FEDERAL

PPP-C-843

MILITARY

MIL-E-10062	-	Engines: Preparation for Shipment and Storage of.
MIL-M-38784	-	Manuals, Technical: General Style and Format Requirements.
MIL-M-63036	-	Manuals, Technical: Operator's, Preparation of.
MIL-M-63038	-	Manuals, Technical: Organizational or Aviation Unit, Direct Support or Aviation Intermediate, and General Support.
MIL-P-116	-	Preservation-Packaging, Method of.
MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-S-12651	-	Sprayers, Packaging of.
MIL-T-704	-	Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-171	-	Finishing of Metal and Wood Surfaces.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1474	-	Noise Limits for Army Materials.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and draft manual form a part of this specification to the extent specified herein. Unless otherwise specified, the drawings and manual shall be those in effect on the date of the solicitation.

FAR 27 (DoD Supplement)

Data Requirements

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:01Z
Check the source to verify that this is the current version before use.

2.1.2 (cont'd)

DoD 5000.19-L, Volume II

DoD Acquisition Management Systems and Data Requirements Control List (AMSDL).

ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Reg. No.241-208-AA

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and Amendments.

DRAWINGS

U.S. ARMY BIOMEDICAL RESEARCH AND DEVELOPMENT LABORATORY (USABRDL)

11134FD - Pesticide Dispersal Unit (PDU), Multicapability, Helicopter Slung.

DRAFT MANUAL

U.S. ARMY BIOMEDICAL RESEARCH AND DEVELOPMENT LABORATORY

Draft TM-13 X-XXX-XXX-X - Operator's, Organizational, and Direct Support Maintenance Manual with Repair Parts List: Pesticide Dispersal Unit, Multicapability, Helicopter Slung.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Section IX - Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:01Z
Check the source to verify that this is the current version before use.

2.2 (cont'd)

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 - Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 - Structural Welding Code, Qualification, Section 5

(Application for copies should be addressed to AWS, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

HONDA MOTOR CO., LTD.

HONDA G340 - Shop Manual.

(Application for copies should be addressed to Honda Motor Co., U.S.A., Ltd., 115 Gaither Drive, Morristown, NJ 08057.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J1174 - Operator Ear Sound Level, Measurement Procedure for Small Engine Powered Equipment.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In event of a conflict between the text of this specification and the references cited herein (except the associated detail specifications), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Drawings. The drawings, 11134 FD, (see 2.1.2) forming a part of this specification are end product drawings. No deviation from the prescribed

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:01Z
Check the source to verify that this is the currentversion before use.

3.1 (cont'd)

dimensions or tolerances is permissible without prior approval of the prescribed on the drawings to ensure correct fit, assembly, and operation of the production item. Unless otherwise specified in the contract or purchase order, any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available for inspection by the contracting officer or his designated representative (see 6.2). In the event of any conflict between the requirements of this specification and the drawings, the later shall govern.

3.2 First article. When specified in the contract or purchase order, three PDUs shall be subjected to first article inspection (see 4.3, 6.2, and 6.4). Examinations and tests shall be as specified herein and during the time frame specified; and unless otherwise specified, all examinations and tests shall be conducted by the Government.

3.3 Identical items. All PDUs furnished under a specific contract shall be physically and mechanically identical; all parts, components, accessories, and subassemblies shall be interchangeable on all PDUs (see 4.5.1 and 4.6). No deviation shall be acceptable without prior written approval from the contracting officer.

3.4 Material. The material shall be as specified herein and on the applicable drawings (see 4.5.1 and 6.6).

3.4.1 Material deterioration and control. The PDU shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against various forms of corrosion and deterioration that may be encountered in any part of the applicable storage and operating environment to which the item may be exposed (see 4.5.1).

3.4.2 Identification of materials and finishes. Unless otherwise specified in the contract or purchase order, the contractor shall identify the specific material, material finish, or treatment for use with components and sub-components, and shall make information available to the contracting officer or designated representative (see 6.2).

3.4.3 Metal fabrication. Metal used in fabrication shall be free from kinks, splitting, buckling, wrinkling, and sharp bends. The straightening of material shall be done by methods that will not cause damage to the metal. All burrs and rough edges shall be removed to provide a smooth surface. All bends of a major character shall be made with dies of fixtures to ensure uniformity of size and shape (see 4.5.1).

3.4.4 Fiberglass fabrication. Fiberglass components shall be free from burrs, cracks, and fractures. Parts bolted to or molded into the fiberglass shall fit tight and not permit leakage of liquid or solid material from the pesticide tank (see 4.5.1).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:01Z
Check the source to verify that this is the currentversion before use.

3.4.5 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with drawings; and all bolts, nuts, and screws shall be torqued properly, consistent with the type of fastening system (see 4.5.1).

3.4.6 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the dispersal unit to conform to the examination and test requirements of section 4.

3.4.6.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Structural Welding Code, AWS D1.1, Qualification, Section 5.

Welding Qualifications of the ASME, Section IX.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedures are equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes (see 4.5.1).

3.4.7 Treatment and painting. The portion of the PDU normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A (see 4.5.1).

3.4.7.1 Color. Unless otherwise specified in the contract or purchase order, all surfaces shall be painted with a paint system conforming to the forest green (383) color in accordance with MIL-STD-171, Color for Army.

3.4.8 Identification markings. The PDU shall be identified in accordance with MIL-STD-130 (see 4.5.1). Plates conforming to MIL-P-514 shall be used for this purpose when specified in section 6.2.

3.5 Performance. The PDU shall perform as follows:

a. Operate continuously (except for refueling) in each operational mode for the time specified in section 4.5.2.3 using the materials specified in section 4.5.2.2.

b. Comply with U.S. Environmental Protection Agency (EPA) requirements for dispensing registered pesticides labeled for pest control as

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

3.5 (cont'd)

well as the Occupational Safety and Health Administration (OSHA) requirements for protection of personnel operating the PDU.

3.5.1 Flow rate. Flow rates for applications in liquid modes shall vary no more than ± 10 percent from the manufacturer's published flow rates when tested in accordance with 4.5.2.3. Variation in flow rates of solid material between chutes shall not exceed 0.5 pound (226.78 grams) when tested in accordance with 4.5.2.4.

3.5.2 ULV droplet size. The dispersal unit shall be capable of rotating the nozzle(s) at the rate specified (see 4.5.2.5) along with producing flow rates and droplet sizes in accordance with the insecticide directions for Cythion® 1/, EPA Reg. No.241-208-AA. Droplets will be collected in accordance with the Cythion directions and as specified in 4.5.2.5.

3.6 Noise limits.

3.6.1 Personnel noise. The noise level at the operator's station when the PDU is operated on the ground shall not exceed 85 decibels (dB(A)) when tested in accordance with 4.5.2.6.1.

3.6.2 Precautions (noise). Hearing caution signs shall be posted on equipment if the sound level exceeds 85 dB(A) at the operator's station when the dispersal unit is operated on the ground. Noise level of 85 dB(A) may only be exceeded when it is beyond the present state-of-the-art. Caution signs, if required, shall state "CAUTION, HEARING PROTECTION REQUIRED WITHIN (SPECIFY) FEET". Specify the maximum distance in feet that 85 dB(A) are measured from the equipment. Plates shall be in accordance with MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material.

3.6.3 Exterior noise. The PDU's noise level shall comply with MIL-STD-1474 when tested in accordance with 4.5.2.6.2.

3.7 Reliability. The minimum acceptable value for mean-time-between-failure (MTBF) shall be 100 hours when tested as specified in 4.5.2.7.

3.8 Maintenance Ratio. The PDU shall have a maintenance ratio of not more than 0.05. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing (see 4.5.2.9).

1/ Registered trademark of American Cyanamid Co. Information on availability of the material can be obtained from the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

3.9 Environmental evaluation. The PDU shall withstand without damage the high temperature storage test, the rain test, and the transportation vibration test (see 4.5.2.9).

3.10 Manuals. The contractor shall provide the following manuals in accordance with 3.10.1 and 3.10.2.

3.10.1 PDU Manual. The Government draft manual TM-13 X-XXX-XXX-X forms a part of this specification to the extent specified herein. The draft manual shall be provided to the contractor. The contractor shall use the Government draft manual as a reference and guide to prepare a type 13-P Operator's, Organizational, and Direct Support Maintenance Manual with Repair Parts List: Pesticide Dispersal Unit, Multicapability, Helicopter Slung Manual in accordance with MIL-M-63036, MIL-M-63038, and MIL-M-38784. The reading level for the manual shall be 7th grade ± 1 year (see 4.5.1). In the event of any conflict between the requirements of this specification, drawings, and the draft manual, the drawings and specification shall govern. The contractor shall provide a verified manual for each PDU produced.

3.10.2 Engine Manual. A commercial operator and repair manual (shop manual) for the engine specified in the Honda G340 manual or equal shall be provided by the contractor with each unit produced.

3.11 Optional items. When required (see 6.2), the following high mortality items shall be specified as spare parts:

a. Air Filter.

b. Spark Plug.

c. Contact Point.

d. Condenser.

e. Fuses.

f. Flow Control Valve O-rings.

g. Flow Control Valve Gaskets.

h. Nylon and Brass Nozzles.

i. Orifices.

j. Cores.

k. Diaphragms.

l. Rotary Atomizer Heads.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:01Z
Check the source to verify that this is the current version before use.

3.11 (cont'd)

m. Rotary Atomizer Seals and Gaskets.

n. Polyethylene Sleeves.

o. Strainers (liquid pesticide and hydraulic).

3.12 Workmanship. The PDU and its various components and connections shall not leak and shall be clean and free from sand, dirt, fins, pits, scale, rust, burrs, sharp edges, cracks, unsound welds, or any other defects that could impair performance or serviceability (see 4.5.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The supplier is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.3 Disassembly inspection. Failure of any test or examination shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and reference drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the supplier.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:01Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. Inspection shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection comparison (see 4.6).

d. Inspection of preparation for delivery (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article unit shall be examined as specified in 4.5.1. Failure of any test shall be cause for rejection.

4.3.2 Tests. The first article unit shall be tested as specified in 4.5.2. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. Acceptable Quality Level (AQL) shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Test. Each PDU shall be tested as specified in 4.5.2.3 except that dispersal time in each mode (LV, ULV, SD) shall be 5 minutes in lieu of 15 minutes.

4.5 Inspection procedure.

4.5.1 Examination. The PDU shall be examined as specified herein for the following defects:

Major

101. Dimensions not as shown on the drawings.
102. Assembly not complete or not correct.
103. Parts or components broken or missing.
104 Materials not as specified.
105. Treatment and painting not as specified.
106. All parts, components, accessories, and subassemblies are not interchangeable on all PDUs.
107. Technical manuals missing or not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

4.5.1 (cont'd)

108. Workmanship not as specified.
109. Leakage of any component.

Minor

110. Accessories or identification marking missing or not as specified.
111. Repair parts missing or not as specified.

4.5.2 Tests.

4.5.2.1 Test preparation. Prior to the start of testing, manuals (Operator's, Organizational, and Direct Support Maintenance Manual with Repair Parts List: Pesticide Dispersal Unit, Multicapability, Helicopter Slung Manual and the commercial operator and repair manual (shop manual) for the engine specified in section 2.2) shall be provided by the contractor. In addition, high mortality spare parts shall be provided. Preventive maintenance checks and services shall also be completed prior to any testing in accordance with the PDU's Operator's Manual.

4.5.2.2 Test materials. Unless otherwise specified the test liquid shall be water and the test solid shall be Celatom® 2/ or equivalent.

4.5.2.3 Performance. With the PDU on the ground, it shall perform as follows:

a. While prepared in the LV mode, the PDU shall disperse water through all available nozzles for a period of not less than 15 minutes at each of the following boom pressures: 207, 279, 349 kilo pascals (kPa) [30, 40, and 50 pounds force per square inch (lbf/square in)]. Any leaks, inability to pressurize the hydraulic or pesticide pump systems, inability to control liquid dispersal on-off functions, or malfunction of any component shall constitute failure of this test.

b. While prepared in the ULV mode, the PDU shall disperse water through each set of nozzles and rotary atomizers for a period of not less than 15 minutes at each of the following boom pressures: 207, 279, 349 kPa (30, 40, and 50 lbf/square in). Any leaks, inability to pressurize the hydraulic or pesticide pump systems, inability to control liquid dispersal on-off functions, or malfunction of any component shall constitute failure of this test.

c. While prepared in the SD mode, the PDU shall operate for a period of not less than 15 minutes dispensing 50 pounds [22.68 kilograms (kg)] of the solid test material through each of the two chutes. The

2/ Registered trademark of Eagle-Picher Industries, Inc. Information on availability of the material can be obtained from the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

4.5.2.3 (cont'd)

metering plates shall be adjusted to the 2.0 position, and the flow control gates shall be opened and closed a minimum of 10 times during dispersal. Any leaks, inability to pressurize the hydraulic system, inability to control solid dispersal on-off functions, or malfunction of any component shall constitute failure of this test.

4.5.2.4 Flow rate. The test materials shall be as specified (see 4.5.2.2) with preparation and flow rate settings as follow:

a. The PDU shall be prepared in the LV mode using D-4-45 orifices and cores. With the boom pressure set at 207 kPa (30 lbf/square in) and all nozzles operating, liquid shall be collected from the 2nd, 8th, and 14th nozzle from the end of each boom for 30 seconds. The liquid collected from all nozzle sites shall be combined and the appropriate calculations made to give flow rate in gallons per minute per nozzle. No less than three trials shall be taken to ensure accuracy.

b. The PDU shall be prepared in the ULV mode using D-4 orifices in the nozzles. With the pesticide line disconnected from the rotary nozzle and the boom pressure set at 207 kPa (30 lbf/square in), the liquid shall be collected from both nozzle sites for 30 seconds. The liquid collected from all nozzle sites shall be combined and the appropriate calculations made to give flow rate in gallons per minute per nozzle. No less than three trials shall be taken to ensure accuracy.

c. The PDU shall be prepared in the SD mode with the exception of not installing the slinger assembly. With the metering plate setting at 2, collect material from both chutes for 30 seconds. Record the weight of material collected from each chute. The above procedure shall be repeated for the settings of 4, 6, and 8. No less than three trials at each setting will be taken to ensure accuracy.

When adjusted by the manufacturer's instructions to disperse liquid at the settings specified above, the actual flow rate shall vary no more than ± 10 percent of the manufacturer's projected flow rate. Variation in flow rates of solid material between chutes shall not exceed 0.5 pound (22.68 kilograms) at any setting.

4.5.2.5 ULV droplet size. The test shall be conducted with a malathion simulant consisting of three parts glycerin to one part water as the spray material. The following equipment conditions shall be established:

a. The rotary atomizers shall be rotated at no less than 10,500 revolutions per minute when under no load.

b. A high density polyethylene 20 micron sleeve will be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

4.5.2.5 (cont'd)

c. A flow rate of 0.5 gallon per minute (1.89 liters per minute) per nozzle shall be established.

d. The spray formulation must be prefiltered through a 10 micron filter prior to transfer into the dispersal unit's tank.

e. The nozzle shall be oriented parallel to the ground.

f. Droplets shall be collected while the sprayer is on the ground.

g. Droplets shall be collected on Teflon® 3/ coated microscope slides. The slides shall be vertically waved once, as rapidly as possible, through the aerosol cloud at a distance of 4 feet (1.22 meters) from the point of discharge.

h. Two slides shall be used to collect droplets. At least 200 droplets per slide shall be collected.

i. Droplets on both slides shall have a volume median diameter (VMD) of less than 30 microns, with no more than 5 percent of the droplets exceeding 75 microns.

j. A spread factor of 0.7 shall be used for the spray formulation.

4.5.2.6 Noise level measurement.

4.5.2.6.1 Personal noise testing. The dispersal unit shall be tested in accordance with the provisions of MIL-STD-1474 for steady-state noise at personnel-occupied areas. Noise levels shall be taken while the system is at maximum rated output. The data should include dB(A) levels taken at the operator's station while running the dispersal unit on the ground.

4.5.2.6.2 Exterior noise testing. The dispersal unit shall be tested in accordance with the provisions for exterior noise of MIL-STD-1474. Test procedure SAE J1174 may be used as an alternative procedure.

4.5.2.6.3 Hazard precautions. Should the noise level exceed 85 dB(A) at any operator position and the excess noise level can be documented as being clearly beyond the state-of-the-art, failure to provide the precautions specified in 3.6.2 shall constitute failure of this test.

3/ Registered trademark of E. I. duPont de Nemours & Co., Inc. Information on availability of slides can be obtained from the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:01Z
Check the source to verify that this is the current version before use.

4.5.2.7 Reliability. Manuals, high mortality spare parts, and common hand tools shall be furnished prior to the start of any testing. The minimum acceptable reliability performance is two or less minor failures during 200 hours of reliability testing. The 200 hours shall consist of 75 hours in the LV mode, 75 hours in the ULV mode and 50 hours in the SD mode. If three or fewer failures occur, the unit shall be considered acceptable. In addition to the overall limit of three failures, each test machine (in accordance with the definitions that follow) shall not experience: (a) any catastrophic failures, (b) more than one major failure, (c) more than three minor failures. Note that a major failure cascades into the minor failure category; that is, a major failure would reduce the allowed number of minor failures by one. The Government may require the manufacturer to make minor modifications to correct the cause of any failure identified in the reliability testing, or the manufacturer may volunteer modifications to prevent failure reoccurrence during testing. Any such modifications are subject to Government approval and shall be accomplished at no cost to the Government. After completion of such modifications, the Government shall determine if any tests need to be repeated or extended. After successful corrective action has been completed, at the discretion of the Government, failures may be eliminated from the count if there is convincing evidence that the cause of the failure has been eliminated. The following definitions apply to reliability failure criteria:

a. General definition for failure - Any malfunction that causes inability to commence operation, cessation of operation, degradation below the operating conditions required to continue the test, or a safety hazard to personnel.

b. Catastrophic failure - A serious failure that totally disables the machine and cannot be repaired in less that 8 man-hours or requires part replacements that exceed 25 percent of the cost of the machine.

c. Major failure - A failure that requires 1 to 8 man-hours to repair or requires part replacements that cost between 3 percent and 25 percent of the machine cost. Sophisticated repair equipment is permitted in this category.

d. Minor failure - A failure that can be corrected in less that 1 man-hour without the use of sophisticated equipment and requires part replacements that do not exceed 3 percent of machine cost.

e. Sophisticated repair equipment - Machine shop type support such as welding, cylinder boring, lathe work, special riveting, etc.

f. Machine cost - The Government's cost per unit.

g. Parts cost - The Government's cost at time of failure.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:01Z
Check the source to verify that this is the current version before use.

4.5.2.8 Maintenance support evaluation. Maintenance operations required during testing shall be accomplished to determine conformance to 3.8. Failure of the support elements to maintain the PDU as specified herein shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the mean time between failures (MTBF).

4.5.2.9 Environmental tests.

4.5.2.9.1 High temperature storage. The high temperature storage test shall be conducted in accordance with MIL-STD-810, method 501.2, procedure I, climate category hot.

4.5.2.9.2 Rain test. The rain test shall be conducted in accordance with MIL-STD-810, method 506.2-11, procedure II (Drip).

4.5.2.9.3 Transportation vibration. The transportation vibration test shall be conducted in accordance with MIL-STD-810, method 514.3, procedure II, category 3 (Loose Cargo Bounce).

4.6 Inspection comparison. The Government may select PDUs at any time during the contract production period and subject these PDUs to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirement of this specification. The inspection will be performed by the Government, at a site selected by the Government, on PDUs selected at random from those which have been accepted by the Government and will not include the previously inspected first article PDUs. In addition to any test specified as part of the inspection comparisons, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison; and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison PDU to meet any requirement specified herein during and as a result of the examination and test specified in 4.5 shall be cause for rejection of the inspection comparison PDU and shall be cause for refusal by the Government to continue acceptance of production PDUs until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on PDUs previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered *prima facie* evidence that all PDUs accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of preparation for delivery. The preservation, packaging, packing, marking, and quality conformance of each PDU shall be inspected in accordance with the requirements of MIL-S-12651. Unless otherwise specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

4.7 (cont'd)

(see 6.2), a preproduction pack shall be furnished within the time frame required for examination and testing in accordance with MIL-S-12651.

4.8 Verification of PDU manual. The Operator's, Organizational, and Direct Support Maintenance Manual with Repair Parts List: Pesticide Dispersal Unit, Multicapability, Helicopter Slung shall be verified by the Government for accuracy and style according to MIL-M-63036, MIL-M-38784, and MIL-M-63038.

5. PACKAGING

5.1 Preservation. Preservation shall be as follows for levels A, B, and C.

5.1.1 Disassembly. The PDU's subassemblies and/or components shall be disassembled to the following level:

a. Bucket assembly - the booms and fin shall be removed, but the liquid cross-tube assembly shall remain installed.

b. Booms - LV booms and struts shall be completely disassembled with nozzles, gauges, and quick-release pins remaining intact; the ULV boom and its components shall remain completely intact.

c. SD components - the slinger assembly and chutes, if installed, shall be removed and the liquid cross-tube assembly installed.

Removed bolts, nuts, pins, screws, and washers shall be reinstalled to mating parts and secured to prevent their loss.

5.1.2 Cleaning. Each PDU shall be cleaned according to section 3.3.1 of MIL-P-116.

5.1.3 Drying. Immediately after cleaning, items shall be dried using applicable procedure in section 3.4 of MIL-P-116.

5.1.4 Preservative application. Preservatives shall not be used.

5.1.5 Engines. The gasoline engine shall be preserved in accordance with level A requirements for type II, Method I of MIL-E-10062.

5.1.6 Batteries and electrolyte. Preservation shall be provided in accordance with section 3.11.1.1.15 of MIL-E-10062.

5.1.7 Hydraulic system. The hydraulic reservoir, lines, and manifolds shall be drained. Flushing with solvent shall not be done. All quick-release fittings shall be disconnected and capped.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

5.1.8 LV and ULV booms. Cushioning material conforming to PPP-C-843, type II, class A, shall be provided as necessary in a manner and in sufficient quantity to protect nozzles, rotary atomizers, and pressure gauges and immobilize the booms and struts.

5.1.9 Other components The remote control assembly, repair parts, maintenance tools, and technical publications shall be preserved according to MIL-S-12651 with cushioning material conforming to PPP-C-843, type II, class A, provided as necessary in a manner and in sufficient quantity to immobilize the components.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. PDUs preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-S-12651 for boxed shipments.

5.2.1.1 Consolidation and arrangement of components. The PDU as disassembled in 5.1.1 shall have its components arranged into three appropriately sized boxes as follows:

a. Box 1 - shall contain bucket assembly, fin, dolly, and battery with electrolyte.

b. Box 2 - shall contain LV booms and struts, and ULV boom.

c. Box 3 - shall contain SD components and other components as stated in 5.1.9.

5.2.2 Level B. Sprayers preserved as specified in 5.1 and consolidated as in 5.2.1.1 shall be packed in accordance with the level B packing requirements of MIL-S-12651.

5.2.3 Level C. Sprayers preserved as specified in 5.1 and consolidated as in 5.2.1.1 shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Level C. Level C marking shall be in accordance with the requirements of ASTM D 3951.

5.4 Workmanship. Workmanship shall be such as to provide maximum protection to the PDU by the application of the requirements specified herein, to prevent corrosion, deterioration, and mechanical or physical damage during handling, shipment, and storage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The PDU is used for dispersing both liquid and solid pesticides from helicopters for control of vector and pest arthropods.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual document referenced (see 2.2).

c. Time frame required for submission of first article (see 3.2).

d. Type, size, class, and composition of identification plates (see 3.4.9).

e. When repair parts shall be required (see 3.11).

f. When first article inspection is required and the number of PDUs to be inspected (sec 4.4).

g. When and where the Government will conduct test and examinations (see 4.6).

h. Level of preservation, packaging, and marking required (see 5.1-5.4).

i. Required data will be made available unless otherwise specified (see 3.1).

j. Identification of materials and finishes shall be made available unless otherwise specified (see 3.4.2).

k. When other than color 383 green is required (see 3.4.8.1).

6.3 Consideration of Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

6.3 (cont'd)

Paragraph No.	Data requirement Title	Applicable DID No.	Option
3.10.1	Manuals, Technical: Operator's Preparation of	DI-M-1517A	18
3.10.1	Manuals, Technical: Organizational, Direct Support, and General Support Maintenance	DI-M-1517A	19
3.10.2	Manuals, Technical: Commercial Equipment	DI-M-6153	N/A

(Data Item Descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of Data Item Descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 Inspection for First Article. When a first article inspection is required, the PDU should be a first article sample. The first article shall consist of three PDUs. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government; and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Disposability. One or more of the following methods shall be used to accomplish disposal of the packaging material: reuse, recycling, sanitary landfill, incineration, or pyrolysis.

6.6 Recycled material. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new, and shall be fabricated using materials produced from recycled materials to the maximum extent practicable without jeopardizing the intended use. The term "recycled materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically authorized (see 3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

6.7 Subject term (key word) listing. The following are a listing of subject items which allow identification of this document during retrieval searches:

Aerosol sprayer
Helicopter slung sprayer
Insecticide sprayer
Military specification, sprayer
Pesticide Dispersal Unit
Sprayer

Custodians:
Army - MD
Navy - YD
Air Force - 99

Review activities:
Army - CE
Navy - MS,CG
Air Force - 84
DLA - CS
AFPMB - PM

User activity:

Preparing activity:
Army - MD

(Project 3740-0123)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

U.S. Army Biomedical Research and
Development Laboratory
ATTN: SGRD-UBE-D
Building 568, Fort Detrick
Frederick, MD 21701-5010

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:01Z
Check the source to verify that this is the current version before use.